



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Experiment No. 1

1. Experiment Title: Demonstrate Inline, Internal, External JavaScript, use console methods; create a webpage showing user info and a welcome message.

2. Software/Tools Required: VS Code, Html/JS Compiler...

3. Experiment Program Code:

Internal:

```
<!DOCTYPE html>

<html>

<head>

    <title>Hello World</title>


<style>

    body{

        background-color:#ffe4e1;

        text-align:center;

        font-family:Arial, sans-serif;

        padding:40px;

    }


    h1{

        font-size:48px;

        color:white;

        background-color:#91ff00;

        padding:15px;

        border-radius:8px;

    }


    p{
```



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```
font-size:20px;

color:#333333;

background-color:#ffffcc;

padding:10px;

margin-top:20px;

}

button{

font-size:18px;

color:white;

background-color:green;

padding:12px 24px;

border:none;

border-radius:5px;

cursor:pointer;

margin-top:20px;

}

</style>

</head>

<body>

<h1>Hello World</h1>

<p>Name : Aaryan K. Kawale , PRN : 24070521162</p>

<p>This is a sample Webpage made using External css.</p>

<button>Click Me</button>

</body>

</html>
```



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External:

```
<!DOCTYPE html>

<html>

<head>

  <title>Hello World</title>

  <link rel="stylesheet" href="style.css">

</head>

<body>


  <h1>Hello World</h1>

  <p> Name : Aaryan Kawale , PRN: 24070521162</p>

  <p> Sec: C, Sem: 5th </p>


  <p>This is a sample Webpage is made by using the External CSS.</p>


  <button>Click Me</button>


</body>

</html>
```

Inline:

```
<!DOCTYPE html>

<html>

<head>

  <title>Hello World</title>

</head>

<body style="background-color:#f0f8ff; text-align:center; font-family:Arial, sans-serif;
padding:40px;">
```



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```
<h1 style="font-size:48px; color:white; background-color:#007BFF; padding:15px;
border-radius:8px;">
```

Hello World

```
</h1>
```

```
<p>
```

Name : Aaryan K. Kawale ,PRN : 24070521162

```
</p>
```

```
<p style="font-size:20px; color:#333333; background-color:#ffffcc; padding:10px;
margin-top:20px;">
```

This is a sample webpage designed by Inline CSS.

```
</p>
```

```
<button style="font-size:18px; color:white; background-color:green; padding:12px
24px; border:none; border-radius:5px; cursor:pointer; margin-top:20px;">
```

Click Me

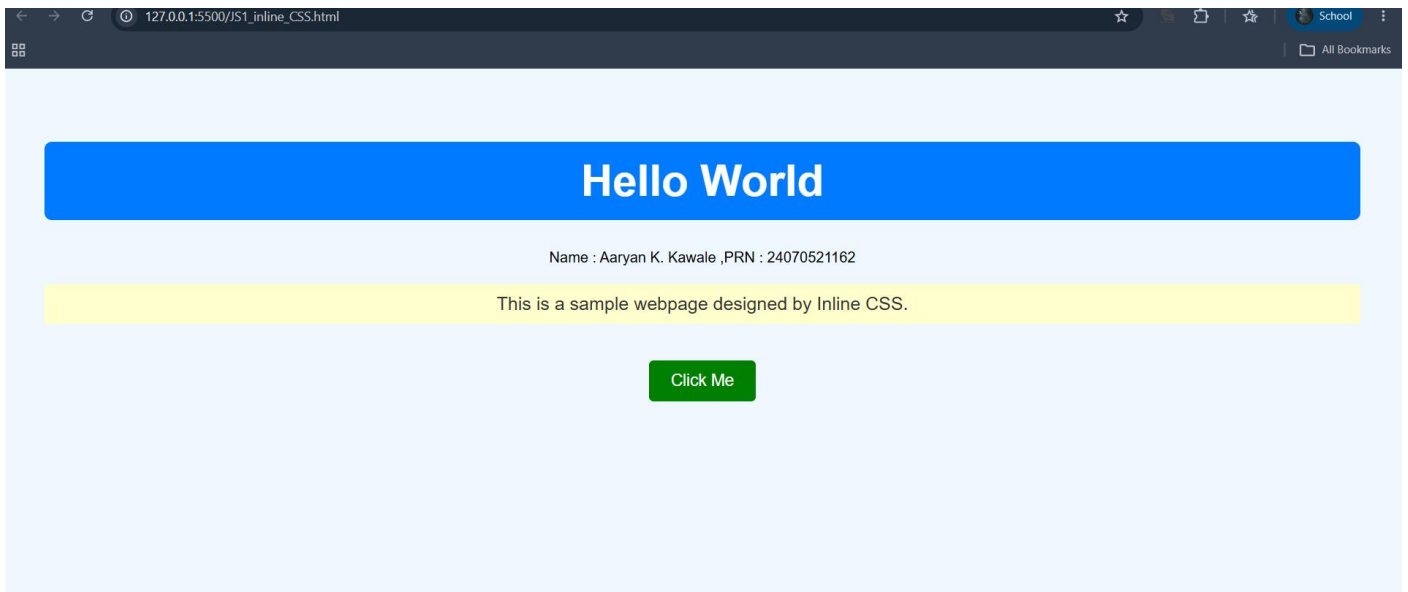
```
</button>
```

```
</body>
```

```
</html>
```

4. Output:

Inline:





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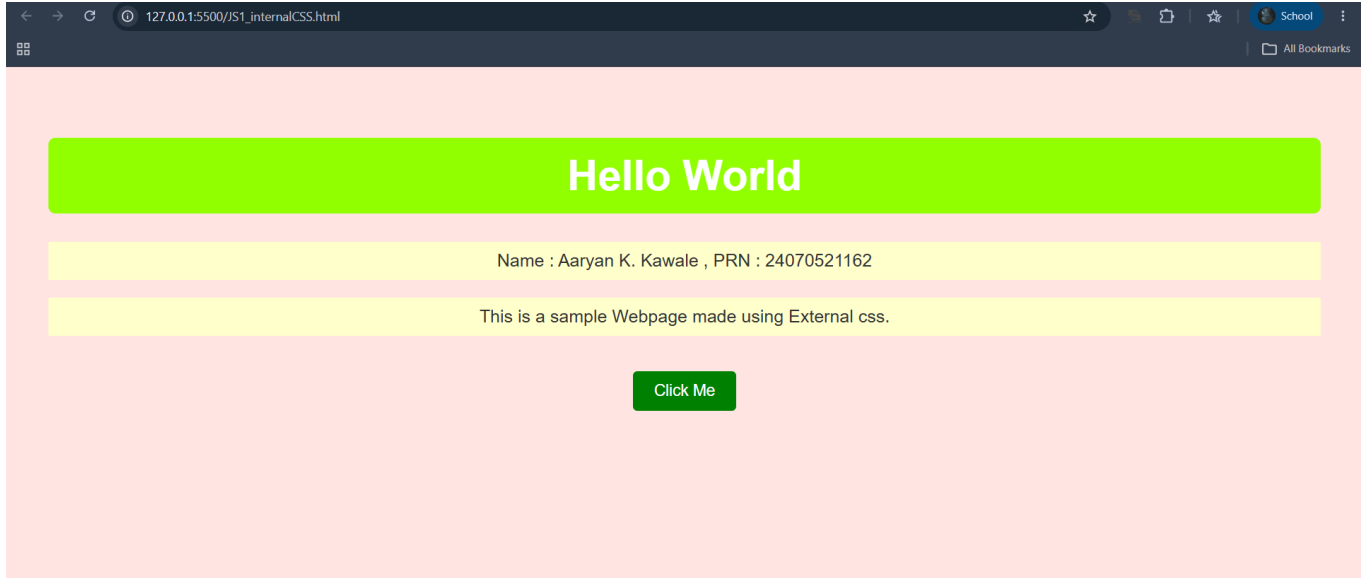
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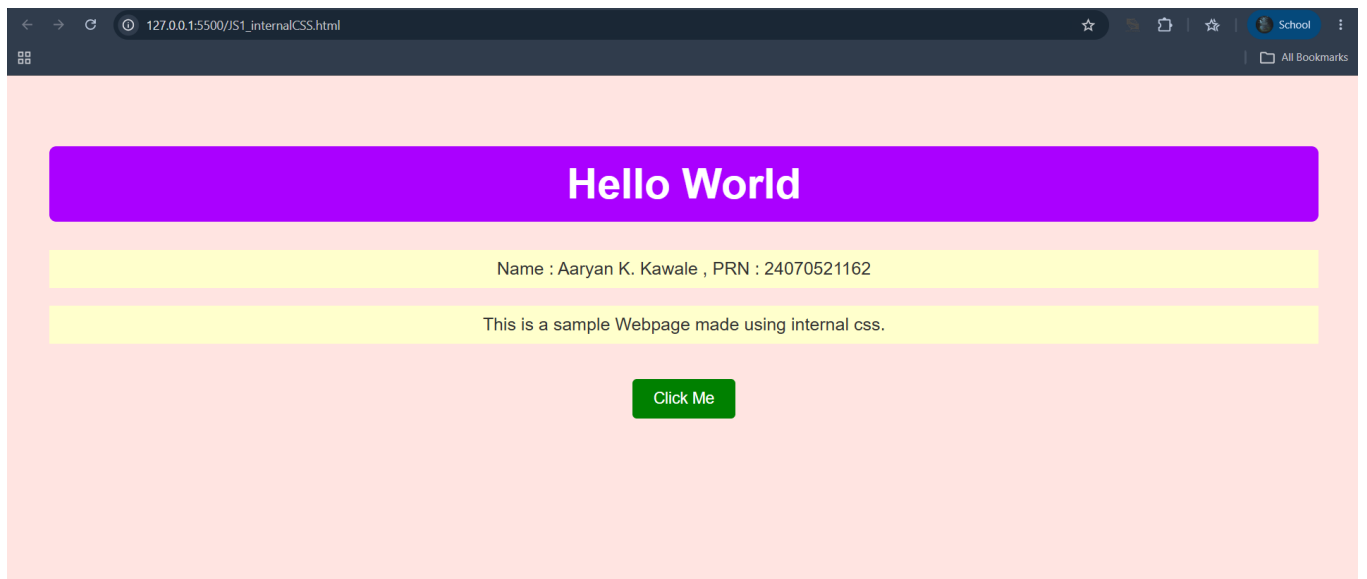
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External:



Internal:





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Case Study 1

1. **Case Study Title:** Desing a webpage for SIT Nagpur with buttons , text , images, etc...

2. **Case Study Program Code:**

```
3. <!DOCTYPE html>
4. <html lang="en">
5. <head>
6.     <meta charset="UTF-8">
7.     <meta name="viewport" content="width=device-width, initial-
scale=1.0">
8.     <title>SIT Nagpur - Computer Science Department</title>
9.     <style>
10.         body {
11.             font-family: Arial, sans-serif;
12.             margin: 0;
13.             padding: 0;
14.             /* Live campus background image */
15.             background-image:
url('https://d3lzc6mbbadaf.cloudfront.net/media/details/ANI-
20230411114128.jpg');
16.             background-size: cover;
17.             background-position: center;
18.             background-attachment: fixed;
19.             background-repeat: no-repeat;
20.         }
21.         header {
22.             background-color: #ae1717b8; /* Symbiosis Maroon */
23.             color: white;
24.             padding: 20px;
25.             text-align: center;
26.         }
27.         .logo {
28.             max-width: 180px;
29.             height: auto;
30.             margin-bottom: 10px;
31.             background-color: transparent;
32.         }
33.         nav {
34.             background-color: #333;
35.             overflow: hidden;
36.         }
37.         nav a {
38.             float: left;
39.             display: block;
40.             color: white;
41.             text-align: center;
```



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```
42.         padding: 14px 20px;
43.         text-decoration: none;
44.     }
45.     nav a:hover {
46.         background-color: #ddd;
47.         color: black;
48.     }
49.
50.     /* Flexbox wrapper to place sections side by side */
51.     .content-wrapper {
52.         display: flex;
53.         flex-direction: row;
54.         justify-content: center;
55.         align-items: stretch;
56.         gap: 30px;
57.         max-width: 1200px;
58.         margin: 50px auto 100px auto;
59.         padding: 0 20px;
60.     }
61.
62.     /* Shared stylings for both white info blocks */
63.     .info-box {
64.         flex: 1;
65.         padding: 30px;
66.         background: rgba(255, 255, 255, 0.92);
67.         border-radius: 8px;
68.         box-shadow: 0 4px 15px rgba(0,0,0,0.3);
69.     }
70.
71.     h2, h3 {
72.         color: #333;
73.     }
74.     ul {
75.         padding-left: 20px;
76.         line-height: 1.6;
77.     }
78.     li {
79.         margin-bottom: 8px;
80.     }
81.
82.     footer {
83.         text-align: center;
84.         padding: 15px;
85.         background-color: #333;
86.         color: white;
87.         position: fixed;
88.         bottom: 0;
89.         width: 100%;
```



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```
90.     }
91.
92.     /* Responsive design for tablets and mobile viewports */
93.     @media (max-width: 768px) {
94.         .content-wrapper {
95.             flex-direction: column;
96.             margin-top: 20px;
97.         }
98.     }
99. </style>
100. </head>
101. <body>
102.
103.     <header>
104.         
107.         <h1>Department of Computer Science & Engineering</h1>
108.         <p>Symbiosis Institute of Technology, Nagpur</p>
109.     </header>
110.
111.     <nav>
112.         <a href="https://sitnagpur.edu.in/">Home</a>
113.         <a href="student_info.html">Student Information</a>
114.     </nav>
115.
116.     <div class="content-wrapper">
117.
118.         <div class="info-box">
119.             <h2>Welcome to the Department</h2>
120.             <p>Welcome to the Department of CSE at SIT Nagpur.
121.             We are committed to delivering world-class engineering education,
122.             fostering innovation, and preparing our students to lead in the global
123.             technology landscape.</p>
124.
125.             <h3>Key Academic Highlights</h3>
126.             <ul>
127.                 <li>State-of-the-art Research & Computing
128.                 Labs</li>
129.                 <li>Industry-aligned & Integrated
130.                 Curriculum</li>
131.                 <li>Excellent Placement Support & Recruitment
132.                 Cells</li>
133.             </ul>
134.         </div>
135.
136.         <div class="info-box">
137.             <h2>Campus Infrastructure</h2>
```




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```
130.         <p>SIT Nagpur provides premium, modern amenities
              designed to enhance student living, academic research, and holistic
              wellness across the campus footprint.</p>
131.
132.         <h3>Facilities Provided</h3>
133.         <ul>
134.             <li><strong>Academic Blocks:</strong> Smart,
              fully air-conditioned digital classrooms.</li>
135.             <li><strong>Central Library:</strong> Digital
              library ecosystem packed with global e-resources.</li>
136.             <li><strong>Living Amenities:</strong> Wi-Fi
              enabled secure hostels for girls & boys.</li>
137.             <li><strong>Dining Options:</strong> In-house
              food mess alongside the modern "Symbi-Eats" cafeteria.</li>
138.             <li><strong>Sports & Gym:</strong> High-end
              fitness setups, sports complex, and wide athletic grounds.</li>
139.         </ul>
140.     </div>
141.
142. </div>
143.
144.     <footer>
145.         &copy; 2026 Symbiosis Institute of Technology, Nagpur.
              All Rights Reserved.
146.     </footer>
147.
148.     <script src="script.js"></script>
149. </body>
150. </html>
```



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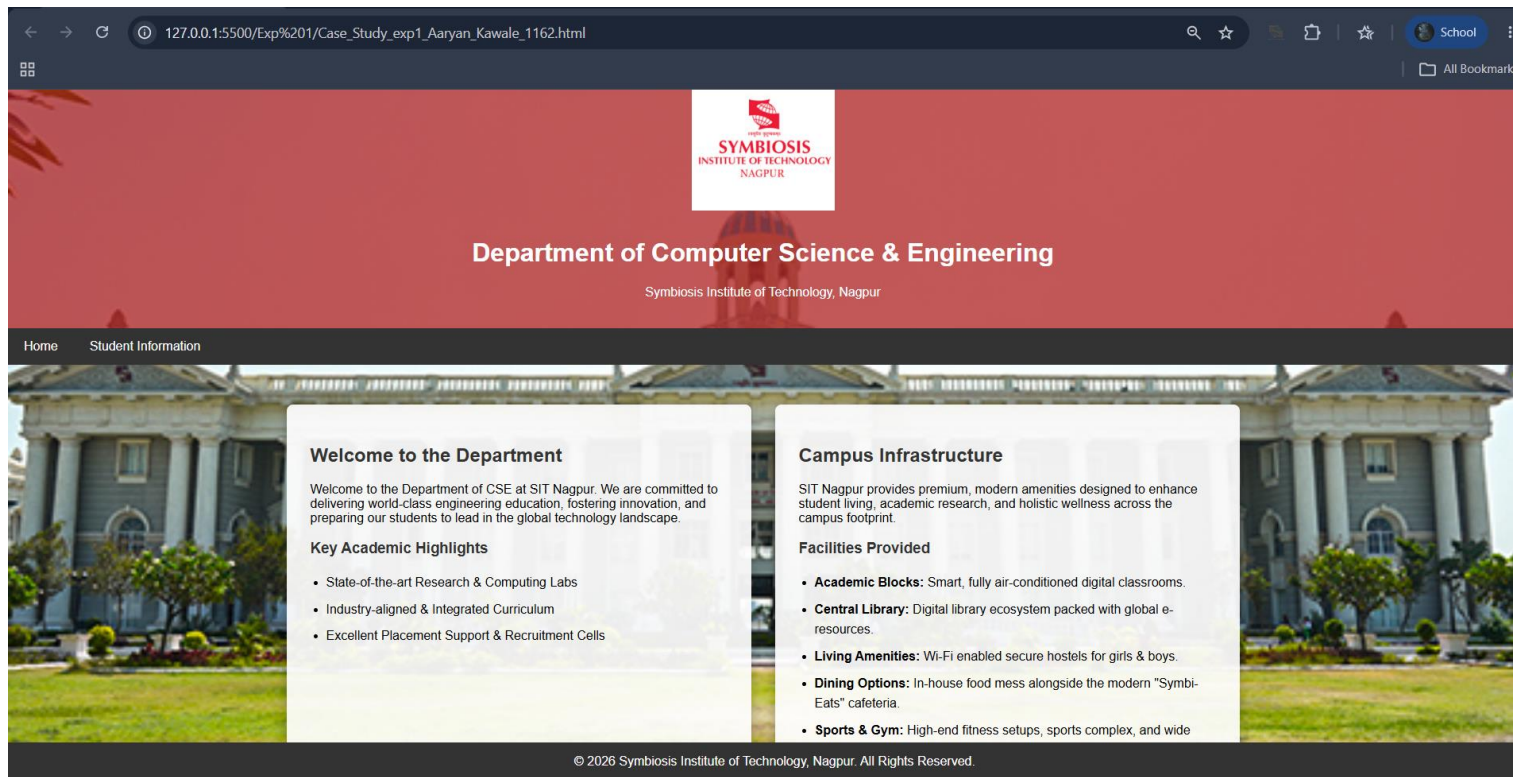
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3.Output:





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Experiment No. 2

- 1. Experiment Title:** Use var, let, const and demonstrate basic features like template literals and destructuring; build a billing calculator with user input.
- 2. Software/Tools Required:** VS Code, Html/JS Compiler...
- 3. Experiment Program Code:**

```
4. <!DOCTYPE html>
5. <html>
6. <head>
7.     <title>Bill Receipt</title>
8.     <style>
9.         body {
10.             background-color: #f4f4f9;
11.             font-family: Arial, sans-serif;
12.             display: flex;
13.             justify-content: center;
14.             align-items: center;
15.             height: 100vh;
16.             margin: 0;
17.         }
18.         .receipt-box {
19.             background-color: #ffffff;
20.             border: 2px solid #333333;
21.             padding: 30px;
22.             width: 350px;
23.             box-shadow: 0 4px 8px rgba(0, 0, 0, 0.1);
24.         }
25.         h3 {
26.             text-align: center;
27.             margin-top: 0;
28.         }
29.     </style>
30. </head>
31. <body>
32.
33.     <script>
34.         // 1. Popups to take user input
35.         var uName = prompt("Enter User Name:");
36.         var contact = prompt("Enter Contact No:");
37.         var pName = prompt("Enter Product Name:");
38.         var qty = Number(prompt("Enter Quantity:"));
39.         var price = Number(prompt("Enter Price per Unit:"));
```



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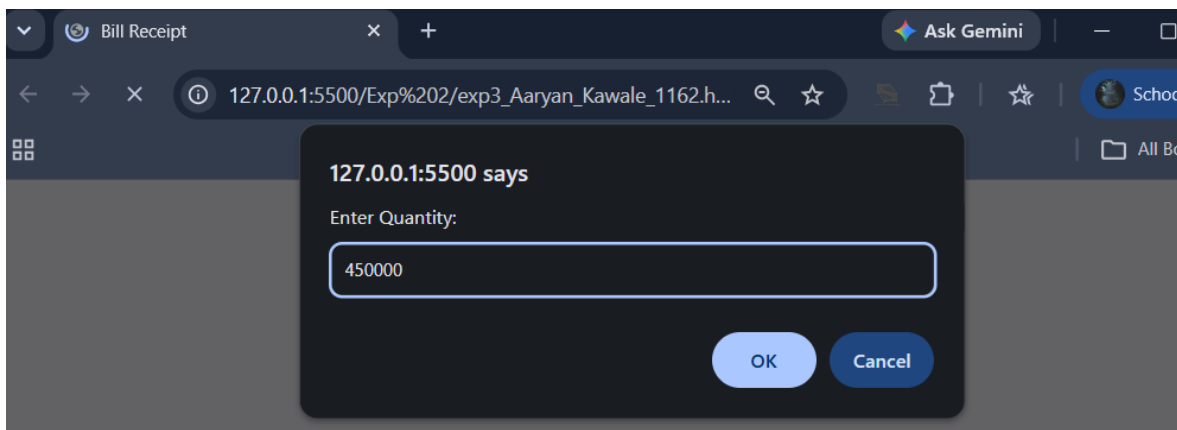
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```
40.
41.     // 2. Mathematical calculations
42.     var total = qty * price;
43.     var gst = total * 0.18;
44.     var finalAmount = total + gst;
45.
46.     // 3. Outputting only the centered bill box
47.     document.write("<div class='receipt-box'>");
48.     document.write("<h3>Bill Receipt</h3>");
49.     document.write("<p><b>User Name:</b> " + uName + "</p>");
50.     document.write("<p><b>Contact No:</b> " + contact + "</p>");
51.     document.write("<p><b>Product Name:</b> " + pName + "</p>");
52.     document.write("<p><b>Quantity:</b> " + qty + "</p>");
53.     document.write("<p><b>Price:</b> " + price + "</p>");
54.     document.write("<hr>");
55.     document.write("<p><b>Total Value:</b> " + total + "</p>");
56.     document.write("<p><b>GST (18%):</b> " + gst + "</p>");
57.     document.write("<p style='font-size: 18px;'><b>Final
Amount:</b> " + finalAmount + "</p>");
58.     document.write("<br>");
59.     document.write("<p style='font-size: 12px; color: #555; text-
align: center;'>Name: Aaryan Kawale, PRN: 24070521162</p>");
60.     document.write("</div>");
61. </script>
62.
63.</body>
64.</html>
```

4.Output:





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Bill Receipt

127.0.0.1:5500 says

Enter Contact No:

24070521162

OK Cancel

127.0.0.1:5500 says

Enter User Name:

Aaryan Kawale

OK Cancel

Bill Receipt

User Name: Aaryan Kawale

Contact No: 24070521162

Product Name: Asus

Quantity: 5

Price: 40000

Total Value: 200000

GST (18%): 36000

Final Amount: 236000

Name: Aaryan Kawale, PRN: 24070521162



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Case Study 2

1. **Case Study Title:** Develop the webpage for an online shopping website using html, css js .

2. Case Study Program Code:

```
3. <!DOCTYPE html>
4. <html lang="en">
5. <head>
6.   <meta charset="UTF-8">
7.   <meta name="viewport" content="width=device-width, initial-
  scale=1.0">
8.   <title>Simple Shop</title>
9.   <style>
10.    * { box-sizing: border-box; margin: 0; padding: 0; font-family:
      sans-serif; }
11.
12.    /* Split layout: Left column for items, right column for cart */
13.    body { display: flex; min-height: 100vh; background: #f4f4f9; }
14.
15.    .products-section { flex: 2; padding: 20px; }
16.    .cart-section { flex: 1; padding: 20px; background: #ffffff;
      border-left: 2px solid #ddd; }
17.
18.    h2 { margin-bottom: 15px; color: #333; }
19.
20.    /* Product Grid */
21.    .grid { display: grid; grid-template-columns: repeat(auto-fill,
      minmax(180px, 1fr)); gap: 15px; }
22.    .card { background: #fff; padding: 15px; border-radius: 8px;
      border: 1px solid #ddd; text-align: center; }
23.    .card h3 { font-size: 1.1rem; margin-bottom: 5px; }
24.    .price { color: #2563eb; font-weight: bold; margin-bottom: 10px; }
25.
26.    button { background: #2563eb; color: white; border: none; padding:
      8px 12px; border-radius: 5px; cursor: pointer; }
27.    button:hover { background: #1d4ed8; }
28.
29.    /* Cart Table & Summary */
30.    table { width: 100%; border-collapse: collapse; margin-bottom:
      20px; }
31.    th, td { text-align: left; padding: 8px; border-bottom: 1px solid
      #eee; }
32.    .remove-btn { background: #ef4444; padding: 4px 8px; font-size:
      0.8rem; }
```




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```
33.     .remove-btn:hover { background: #dc2626; }
34.
35.     .total-box { font-size: 1.2rem; font-weight: bold; display: flex;
36.     justify-content: space-between; margin-top: 15px; }
37.     .checkout-btn { width: 100%; padding: 10px; background: #10b981;
38.     margin-top: 15px; font-size: 1rem; }
39.     .checkout-btn:hover { background: #059669; }
40. </style>
41. </head>
42. <body>
43.     <!-- LEFT COLUMN: ITEMS -->
44.     <section class="products-section">
45.         <h2>Available Items</h2>
46.         <div class="grid" id="productGrid"></div>
47.     </section>
48.     <!-- RIGHT COLUMN: CART & TOTAL -->
49.     <section class="cart-section">
50.         <h2>Shopping Cart</h2>
51.         <table>
52.             <thead>
53.                 <tr>
54.                     <th>Item</th>
55.                     <th>Qty</th>
56.                     <th>Price</th>
57.                 </tr>
58.             </thead>
59.             <tbody id="cartTable"></tbody>
60.         </table>
61.
62.         <div class="total-box">
63.             <span>Total:</span>
64.             <span id="grandTotal">₹0.00</span>
65.         </div>
66.
67.         <button class="checkout-btn"
68.         onclick="checkout()">Checkout</button>
69.     </section>
70.
71.     <script>
72.         // 1. Items Data (Prices in INR)
73.         const products = [
74.             { id: 1, name: "Wireless Mouse", price: 1428.00 },
75.             { id: 2, name: "Keyboard", price: 2856.00 },
76.             { id: 3, name: "Headphones", price: 4284.00 },
77.             { id: 4, name: "USB Cable", price: 761.60 }
```



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```
78.     ];
79.
80.     let cart = [];
81.
82.     // 2. Render Product Cards (Left Column)
83.     function renderProducts() {
84.         const grid = document.getElementById("productGrid");
85.         grid.innerHTML = products.map(p => `
86.             <div class="card">
87.                 <h3>${p.name}</h3>
88.                 <p class="price">₹${p.price.toFixed(2)}</p>
89.                 <button onclick="addToCart(${p.id})">Add to Cart</button>
90.             </div>
91.         `).join('');
92.     }
93.
94.     // 3. Add to Cart Logic
95.     function addToCart(productId) {
96.         const item = cart.find(i => i.id === productId);
97.         if (item) {
98.             item.qty += 1;
99.         } else {
100.             const product = products.find(p => p.id === productId);
101.             cart.push({ ...product, qty: 1 });
102.         }
103.         renderCart();
104.     }
105.
106.     // 4. Remove from Cart Logic
107.     function removeFromCart(productId) {
108.         cart = cart.filter(i => i.id !== productId);
109.         renderCart();
110.     }
111.
112.     // 5. Render Cart Items & Total (Right Column)
113.     function renderCart() {
114.         const table = document.getElementById("cartTable");
115.         const totalEl = document.getElementById("grandTotal");
116.
117.         if (cart.length === 0) {
118.             table.innerHTML = `<tr><td colspan="4" style="text-align:center; color:#888;">Cart is empty</td></tr>`;
119.             totalEl.textContent = "₹0.00";
120.             return;
121.         }
122.
123.         table.innerHTML = cart.map(i => `
124.             <tr>
```




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```
125.         <td>${i.name}</td>
126.         <td>${i.qty}</td>
127.         <td>₹${(i.price * i.qty).toFixed(2)}</td>
128.         <td><button class="remove-btn"
onclick="removeFromCart(${i.id})">X</button></td>
129.     </tr>
130.     `).join('');
131.
132.     const total = cart.reduce((sum, item) => sum +
(item.price * item.qty), 0);
133.     totalEl.textContent = `₹${total.toFixed(2)}`;
134. }
135.
136. function checkout() {
137.     if (cart.length === 0) return alert("Your cart is
empty!");
138.     alert("Order placed successfully!");
139.     cart = [];
140.     renderCart();
141. }
142.
143. // Initial load
144. renderProducts();
145. renderCart();
146. </script>
147.
148. </body>
149. </html>
```

3.Output:

127.0.0.1:5500/Exp%202/Case_Study2_Aaryan_Kawale_1162.html

Available Items

Wireless Mouse
₹1428.00
Add to Cart

Keyboard
₹2856.00
Add to Cart

Headphones
₹4284.00
Add to Cart

USB Cable
₹761.60
Add to Cart

Shopping Cart

Item	Qty	Price	
USB Cable	1	₹761.60	X
Headphones	1	₹4284.00	X
Keyboard	1	₹2856.00	X
Wireless Mouse	1	₹1428.00	X

Total: ₹9329.60

Checkout

Mouza - Wathoda, Nagpur 440008, Maharashtra, India. | Tel: +91-712-6192370 / +91-712-6192368
Email: info@sitnagpur.siu.edu.in | Website: www.sitnagpur.edu.in



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Experiment No. 3

- 1. Experiment Title:** Implement control structures and form validations; create a grading system based on user-entered marks.
- 2. Software/Tools Required:** VS Code, Html/JS Compiler...
- 3. Experiment Program Code:**

```
4. <!DOCTYPE html>
5. <html lang="en">
6. <head>
7.   <meta charset="UTF-8">
8.   <meta name="viewport" content="width=device-width, initial-
  scale=1.0">
9.   <title>5-Subject Grading System</title>
10. <style>
11.   * { box-sizing: border-box; font-family: system-ui, -apple-system,
  sans-serif; }
12.   body { background-color: #f4f6f9; display: flex; justify-content:
  center; align-items: center; min-height: 100vh; margin: 0; padding:
  1.5rem; }
13.   .card { background: #ffffff; padding: 2rem; border-radius: 12px;
  box-shadow: 0 4px 16px rgba(0,0,0,0.08); width: 100%; max-width:
  480px; }
14.   h2 { margin-top: 0; color: #1e293b; text-align: center; }
15.   .grid { display: grid; grid-template-columns: 1fr 1fr; gap:
  0.75rem 1rem; }
16.   .form-group { margin-bottom: 0.75rem; }
17.   .full-width { grid-column: span 2; }
18.   label { display: block; margin-bottom: 0.35rem; font-weight: 600;
  color: #334155; font-size: 0.9rem; }
19.   input[type="number"] { width: 100%; padding: 0.65rem; border: 1px
  solid #cbd5e1; border-radius: 6px; font-size: 0.95rem; }
20.   input[type="number"]:focus { outline: none; border-color: #2563eb;
  box-shadow: 0 0 0 3px rgba(37,99,235,0.15); }
21.   button { width: 100%; padding: 0.75rem; background-color: #2563eb;
  color: white; border: none; border-radius: 6px; font-size: 1rem; font-
  weight: 600; cursor: pointer; margin-top: 0.5rem; transition:
  background 0.2s; }
22.   button:hover { background-color: #1d4ed8; }
23.   .error-banner { background-color: #fef2f2; border-left: 4px solid
  #ef4444; color: #991b1b; padding: 0.75rem; border-radius: 4px; font-
  size: 0.875rem; margin-bottom: 1rem; display: none; }
24.   .result-box { margin-top: 1.25rem; padding: 1.25rem; border-
  radius: 8px; display: none; }
```



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```
25. .result-header { font-size: 1.1rem; font-weight: 700; margin-
    bottom: 0.5rem; text-align: center; }
26. .result-details { font-size: 0.95rem; line-height: 1.5; }
27. /* Dynamic Theme Colors */
28. .grade-A { background-color: #dcfce7; color: #166534; border: 1px
    solid #bbf7d0; }
29. .grade-B { background-color: #e0f2fe; color: #075985; border: 1px
    solid #bae6fd; }
30. .grade-C { background-color: #fef9c3; color: #854d0e; border: 1px
    solid #fef08a; }
31. .grade-D { background-color: #ffedd5; color: #9a3412; border: 1px
    solid #fed7aa; }
32. .grade-F { background-color: #fee2e2; color: #991b1b; border: 1px
    solid #fca5a5; }
33. </style>
34.</head>
35.<body>
36.
37. <div class="card">
38.   <h2>Student Assessment</h2>
39.
40.   <!-- Unified Error Banner -->
41.   <div id="errorBanner" class="error-banner"></div>
42.
43.   <form id="gradeForm" novalidate>
44.     <div class="grid">
45.       <div class="form-group">
46.         <label for="sub1">Mathematics</label>
47.         <input type="number" id="sub1" class="subject-input"
placeholder="0 - 100">
48.       </div>
49.       <div class="form-group">
50.         <label for="sub2">Science</label>
51.         <input type="number" id="sub2" class="subject-input"
placeholder="0 - 100">
52.       </div>
53.       <div class="form-group">
54.         <label for="sub3">English</label>
55.         <input type="number" id="sub3" class="subject-input"
placeholder="0 - 100">
56.       </div>
57.       <div class="form-group">
58.         <label for="sub4">History</label>
59.         <input type="number" id="sub4" class="subject-input"
placeholder="0 - 100">
60.       </div>
61.       <div class="form-group full-width">
62.         <label for="sub5">Computer Science</label>
```



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```
63.     <input type="number" id="sub5" class="subject-input"
placeholder="0 - 100">
64.     </div>
65.     </div>
66.     <button type="submit">Calculate Final Results</button>
67. </form>
68.
69. <div id="resultBox" class="result-box">
70.     <div id="resultHeader" class="result-header"></div>
71.     <div id="resultDetails" class="result-details"></div>
72. </div>
73. </div>
74.
75. <script>
76.     document.getElementById('gradeForm').addEventListener('submit',
function(e) {
77.         e.preventDefault();
78.
79.         const inputs = document.querySelectorAll('.subject-input');
80.         const errorBanner = document.getElementById('errorBanner');
81.         const resultBox = document.getElementById('resultBox');
82.         const resultHeader = document.getElementById('resultHeader');
83.         const resultDetails = document.getElementById('resultDetails');
84.
85.         // Reset UI state
86.         errorBanner.style.display = 'none';
87.         resultBox.style.display = 'none';
88.         resultBox.className = 'result-box';
89.
90.         const scores = [];
91.         let hasFailedSubject = false;
92.
93.         // --- 1. CONTROL STRUCTURE: Loop with Validation Checks ---
94.         for (let i = 0; i < inputs.length; i++) {
95.             const value = inputs[i].value.trim();
96.             const label = inputs[i].previousElementSibling.textContent;
97.
98.             // Check empty input
99.             if (value === '') {
100.                 showError(`Please fill in marks for
<strong>${label}</strong>.`);
101.                 inputs[i].focus();
102.                 return;
103.             }
104.
105.             const score = Number(value);
106.
107.             // Check invalid numbers or out-of-bounds inputs
```



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```
108.         if (isNaN(score) || score < 0 || score > 100) {
109.             showError(`Invalid mark for
    <strong>${label}</strong>. Enter a value between 0 and 100.`);
110.             inputs[i].focus();
111.             return;
112.         }
113.
114.         // Check for subject failure condition (e.g., passing
    mark is 40)
115.         if (score < 40) {
116.             hasFailedSubject = true;
117.         }
118.
119.         scores.push(score);
120.     }
121.
122.     // --- 2. CALCULATIONS ---
123.     const totalMarks = scores.reduce((sum, current) => sum +
    current, 0);
124.     const percentage = (totalMarks / (scores.length * 100)) *
    100;
125.
126.     // --- 3. CONTROL STRUCTURE: Grading Logic ---
127.     let grade = '';
128.     let status = '';
129.     let cssClass = '';
130.
131.     if (hasFailedSubject) {
132.         // Essential academic condition: Failing any individual
    subject overrides average grade
133.         status = 'FAILED (Failed 1 or more subjects)';
134.         grade = 'F';
135.         cssClass = 'grade-F';
136.     } else if (percentage >= 90) {
137.         status = 'PASSED';
138.         grade = 'A+ (Outstanding)';
139.         cssClass = 'grade-A';
140.     } else if (percentage >= 80) {
141.         status = 'PASSED';
142.         grade = 'A (Excellent)';
143.         cssClass = 'grade-A';
144.     } else if (percentage >= 70) {
145.         status = 'PASSED';
146.         grade = 'B (Good)';
147.         cssClass = 'grade-B';
148.     } else if (percentage >= 60) {
149.         status = 'PASSED';
150.         grade = 'C (Satisfactory)';
```



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```
151.         cssClass = 'grade-C';
152.     } else if (percentage >= 50) {
153.         status = 'PASSED';
154.         grade = 'D (Pass)';
155.         cssClass = 'grade-D';
156.     } else {
157.         status = 'FAILED';
158.         grade = 'F (Needs Improvement)';
159.         cssClass = 'grade-F';
160.     }
161.
162.     // --- 4. RENDER OUTPUT ---
163.     resultHeader.textContent = `Status: ${status}`;
164.     resultDetails.innerHTML = `
165.         <strong>Total Marks:</strong> ${totalMarks} / 500<br>
166.         <strong>Overall Percentage:</strong>
167.         ${percentage.toFixed(2)}%<br>
168.         <strong>Final Grade:</strong> ${grade}<br>
169.         <hr style="margin: 0.75rem 0; border: none; border-top:
170.         1px dashed rgba(0,0,0,0.2);">
171.         <strong>Student Name:</strong> Aaryan K. Kawale<br>
172.         <strong>PRN:</strong> 24070521162
173.         `;
174.     resultBox.classList.add(cssClass);
175.     resultBox.style.display = 'block';
176.
177.     function showError(htmlMessage) {
178.         const errorBanner =
179.         document.getElementById('errorBanner');
180.         errorBanner.innerHTML = htmlMessage;
181.         errorBanner.style.display = 'block';
182.     }
183. </script>
184. </body>
185. </html>
```




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4.Output:

Student Grading System

5-Subject Grading System

Ask Gemini

127.0.0.1:5500/Exp%203/exp3.html

School

All Bookmarks

Student Assessment

Mathematics	Science
85	84
English	History
85	74
Computer Science	
98	

Calculate Final Results

Status: PASSED

Total Marks: 426 / 500

Overall Percentage: 85.20%

Final Grade: A (Excellent)

Student Name: Aaryan K. Kawale

PRN: 24070521162



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Case Study 3

1. Case Study Title: Desing a webpage that match password with the user password, if matched it displays result, if not result not displayed.

2. Case Study Program Code:

```
3. <!DOCTYPE html>
4. <html lang="en">
5. <head>
6.   <meta charset="UTF-8">
7.   <meta name="viewport" content="width=device-width, initial-
  scale=1.0">
8.   <title>Result Portal</title>
9.   <style>
10.    * {
11.      box-sizing: border-box;
12.      margin: 0;
13.      padding: 0;
14.      font-family: 'Segoe UI', sans-serif;
15.    }
16.
17.    body {
18.      display: flex;
19.      min-height: 100vh;
20.      background: #f8fafc;
21.    }
22.
23.    /* Split Screen Layout */
24.    .section-left {
25.      flex: 1;
26.      padding: 40px 20px;
27.      background: #ffffff;
28.      border-right: 2px solid #e2e8f0;
29.      display: flex;
30.      justify-content: center;
31.      align-items: center;
32.    }
33.
34.    .section-right {
35.      flex: 1;
36.      padding: 40px 20px;
37.      display: flex;
38.      justify-content: center;
39.      align-items: center;
```



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```
40.     }
41.
42.     /* Form Design */
43.     .box {
44.         width: 100%;
45.         max-width: 320px;
46.     }
47.
48.     h2 {
49.         margin-bottom: 20px;
50.         color: #0f172a;
51.     }
52.
53.     label {
54.         display: block;
55.         font-weight: 600;
56.         margin-bottom: 8px;
57.         color: #334155;
58.     }
59.
60.     input {
61.         width: 100%;
62.         padding: 10px 12px;
63.         border: 1px solid #cbd5e1;
64.         border-radius: 6px;
65.         font-size: 16px;
66.         margin-bottom: 15px;
67.         outline: none;
68.     }
69.
70.     input:focus {
71.         border-color: #2563eb;
72.     }
73.
74.     button {
75.         width: 100%;
76.         padding: 10px;
77.         background: #2563eb;
78.         color: white;
79.         border: none;
80.         border-radius: 6px;
81.         font-size: 16px;
82.         font-weight: 600;
83.         cursor: pointer;
84.     }
85.
86.     button:hover {
87.         background: #1d4ed8;
```



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```
88.     }
89.
90.     .error {
91.         color: #dc2626;
92.         background: #fef2f2;
93.         border: 1px solid #fecaca;
94.         padding: 8px;
95.         border-radius: 6px;
96.         margin-top: 12px;
97.         display: none;
98.         text-align: center;
99.         font-size: 14px;
100.    }
101.
102.    /* Result Card Design */
103.    .placeholder {
104.        color: #64748b;
105.        font-size: 16px;
106.    }
107.
108.    .result-card {
109.        display: none;
110.        background: white;
111.        border: 2px solid #10b981;
112.        border-radius: 12px;
113.        padding: 30px;
114.        text-align: center;
115.        box-shadow: 0 4px 6px -1px rgba(0, 0, 0, 0.1);
116.        width: 260px;
117.    }
118.
119.    .status {
120.        background: #d1fae5;
121.        color: #065f46;
122.        padding: 4px 12px;
123.        border-radius: 12px;
124.        font-weight: bold;
125.        font-size: 14px;
126.        display: inline-block;
127.        margin-bottom: 15px;
128.    }
129.
130.    .gpa {
131.        font-size: 48px;
132.        font-weight: 800;
133.        color: #2563eb;
134.    }
135.
```



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```
136.         .label {
137.             color: #64748b;
138.             font-size: 14px;
139.             font-weight: 600;
140.         }
141.     </style>
142. </head>
143. <body>
144.
145.     <!-- LEFT COLUMN: PASSWORD INPUT -->
146.     <div class="section-left">
147.         <div class="box">
148.             <h2>Student Login</h2>
149.             <form id="loginForm">
150.                 <label for="passInput">Password</label>
151.                 <!-- type="text" की वजह से पासवर्ड टाइप करते वक्त दिखता
रहेगा -->
152.                 <input type="text" id="passInput" placeholder="Enter
password..." required autocomplete="off">
153.                 <button type="submit">Submit</button>
154.             </form>
155.             <div id="errorMsg" class="error">Incorrect
Password!</div>
156.         </div>
157.     </div>
158.
159.     <!-- RIGHT COLUMN: RESULT DISPLAY -->
160.     <div class="section-right">
161.         <div id="placeholder" class="placeholder">🔒 Enter correct
password to view result</div>
162.
163.         <div id="resultCard" class="result-card">
164.             <span class="status">Pass</span>
165.             <div class="gpa">9</div>
166.             <div class="label">GPA</div>
167.         </div>
168.     </div>
169.
170.     <!-- JAVASCRIPT -->
171.     <script>
172.         const TARGET_PASS = "Aaryan1162";
173.
174.         document.getElementById("loginForm").addEventListener("submit", function(e) {
175.             e.preventDefault();
176.             const inputVal =
document.getElementById("passInput").value.trim();
177.
```



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```
178.         const isCorrect = (inputVal === TARGET_PASS);
179.
180.         document.getElementById("errorMsg").style.display =
           isCorrect ? "none" : "block";
181.         document.getElementById("resultCard").style.display =
           isCorrect ? "block" : "none";
182.         document.getElementById("placeholder").style.display =
           isCorrect ? "none" : "block";
183.     });
184. </script>
185.
186. </body>
187. </html>
```

3.Output:

Student Login

Password

Aaryan1162

Submit

Pass

9

GPA

Student Login

Password

Aaryan24070521162

Submit

Incorrect Password!

Enter correct password to view result



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Experiment No. 4

- 1. Experiment Title:** Use Function types, scope, and closures; apply try and catch; build a palindrome checker.
- 2. Software/Tools Required:** VS Code, Html/JS Compiler...
- 3. Experiment Program Code:**

```
4. <!DOCTYPE html>
5. <html lang="en">
6. <head>
7.   <meta charset="UTF-8">
8.   <meta name="viewport" content="width=device-width, initial-
  scale=1.0">
9.   <title>Palindrome Checker</title>
10.  <style>
11.    :root {
12.      --bg-color: #f4f6f8;
13.      --card-bg: #ffffff;
14.      --primary: #4f46e5;
15.      --primary-hover: #4338ca;
16.      --text-main: #1f2937;
17.      --text-muted: #6b7280;
18.      --success-bg: #d1fae5;
19.      --success-text: #065f46;
20.      --error-bg: #fee2e2;
21.      --error-text: #991b1b;
22.    }
23.
24.    * {
25.      box-sizing: border-box;
26.      margin: 0;
27.      padding: 0;
28.      font-family: -apple-system, BlinkMacSystemFont, "Segoe UI",
  Roboto, Helvetica, Arial, sans-serif;
29.    }
30.
31.    body {
32.      background-color: var(--bg-color);
33.      color: var(--text-main);
34.      display: flex;
35.      justify-content: center;
36.      align-items: center;
37.      min-height: 100vh;
```



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```
38.     padding: 20px;
39.   }
40.
41.   .container {
42.     background-color: var(--card-bg);
43.     padding: 32px;
44.     border-radius: 12px;
45.     box-shadow: 0 10px 15px -3px rgba(0, 0, 0, 0.1), 0 4px 6px -2px
rgba(0, 0, 0, 0.05);
46.     width: 100%;
47.     max-width: 480px;
48.   }
49.
50.   h1 {
51.     font-size: 1.5rem;
52.     margin-bottom: 8px;
53.     color: var(--text-main);
54.     text-align: center;
55.   }
56.
57.   p.subtitle {
58.     font-size: 0.9rem;
59.     color: var(--text-muted);
60.     text-align: center;
61.     margin-bottom: 24px;
62.   }
63.
64.   .input-group {
65.     display: flex;
66.     flex-direction: column;
67.     gap: 12px;
68.   }
69.
70.   input[type="text"] {
71.     width: 100%;
72.     padding: 12px 16px;
73.     font-size: 1rem;
74.     border: 2px solid #e5e7eb;
75.     border-radius: 8px;
76.     outline: none;
77.     transition: border-color 0.2s;
78.   }
79.
80.   input[type="text"]:focus {
81.     border-color: var(--primary);
82.   }
83.
84.   button {
```




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```
85.     width: 100%;
86.     padding: 12px;
87.     font-size: 1rem;
88.     font-weight: 600;
89.     color: #ffffff;
90.     background-color: var(--primary);
91.     border: none;
92.     border-radius: 8px;
93.     cursor: pointer;
94.     transition: background-color 0.2s;
95. }
96.
97. button:hover {
98.     background-color: var(--primary-hover);
99. }
100.
101.     #result {
102.         margin-top: 20px;
103.         padding: 12px 16px;
104.         border-radius: 8px;
105.         font-size: 0.95rem;
106.         text-align: center;
107.         display: none;
108.         word-break: break-word;
109.     }
110.
111.     #result.success {
112.         display: block;
113.         background-color: var(--success-bg);
114.         color: var(--success-text);
115.     }
116.
117.     #result.error {
118.         display: block;
119.         background-color: var(--error-bg);
120.         color: var(--error-text);
121.     }
122. </style>
123. </head>
124. <body>
125.
126.     <div class="container">
127.         <h1>Palindrome Checker</h1>
128.         <p class="subtitle">Enter a word, number, or phrase to
129. test</p>
130.         <div class="input-group">
```



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```
131.         <input type="text" id="userInput" placeholder="e.g.,
           racecar, 12321, or A man, a plan..." />
132.         <button onclick="checkPalindrome()">Check</button>
133.     </div>
134.
135.     <div id="result"></div>
136. </div>
137.
138. <script>
139.     function checkPalindrome() {
140.         const inputElement =
           document.getElementById("userInput");
141.         const resultElement = document.getElementById("result");
142.         const rawText = inputElement.value;
143.
144.         if (!rawText.trim()) {
145.             resultElement.className = "error";
146.             resultElement.textContent = "Please enter something to
           check!";
147.             return;
148.         }
149.
150.         // Clean the string: remove non-alphanumeric characters
           and convert to lowercase
151.         const cleaned = rawText.toLowerCase().replace(/[^\a-z0-
           9]/g, "");
152.
153.         // Reverse the cleaned string
154.         const reversed = cleaned.split("").reverse().join("");
155.
156.         // Compare cleaned string with its reversed counterpart
157.         if (cleaned === reversed) {
158.             resultElement.className = "success";
159.             resultElement.innerHTML =
           `<strong>"${rawText}"</strong> is a palindrome!`;
160.         } else {
161.             resultElement.className = "error";
162.             resultElement.innerHTML =
           `<strong>"${rawText}"</strong> is NOT a palindrome.`;
163.         }
164.     }
165.
166.     // Allow pressing "Enter" key to trigger the check
167.     document.getElementById("userInput").addEventListener("keyp
           ress", function(event) {
168.         if (event.key === "Enter") {
169.             checkPalindrome();
170.         }
171.     });
172. }
```



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```
171.     });  
172.     </script>  
173.  
174.     </body>  
175.     </html>
```

4.Output;

Palindrom Checker

Enter a word, number, or phrase to test

Check

"1162-2611" is a palindrome!

Palindrom Checker

Enter a word, number, or phrase to test

Check

"Aaryan" is NOT a palindrome.



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Case Study 4

1. Case Study Title: Desing a webpage to check whether the password is Palindrome or not if Palindrome display msg Weak Pass if not display Strong Pass.

2. Case Study Program Code:

```
3. <!DOCTYPE html>
4. <html lang="en">
5. <head>
6.   <meta charset="UTF-8">
7.   <meta name="viewport" content="width=device-width, initial-
  scale=1.0">
8.   <title>Password Palindrome & Strength Checker</title>
9.   <style>
10.    :root {
11.      --bg-color: #0f172a;
12.      --card-bg: #1e293b;
13.      --primary: #6366f1;
14.      --primary-hover: #4f46e5;
15.      --text-main: #f8fafc;
16.      --text-muted: #94a3b8;
17.      --warning-bg: #451a03;
18.      --warning-text: #fde047;
19.      --warning-border: #854d0e;
20.      --success-bg: #064e3b;
21.      --success-text: #6ee7b7;
22.      --success-border: #047857;
23.      --error-bg: #450a0a;
24.      --error-text: #fca5a5;
25.      --error-border: #991b1b;
26.    }
27.
28.    * {
29.      box-sizing: border-box;
30.      margin: 0;
31.      padding: 0;
32.      font-family: -apple-system, BlinkMacSystemFont, "Segoe UI",
  Roboto, Helvetica, Arial, sans-serif;
33.    }
34.
35.    body {
36.      background-color: var(--bg-color);
37.      color: var(--text-main);
38.      display: flex;
39.      justify-content: center;
```



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```
40.     align-items: center;
41.     min-height: 100vh;
42.     padding: 20px;
43. }
44.
45. .container {
46.     background-color: var(--card-bg);
47.     padding: 32px;
48.     border-radius: 12px;
49.     box-shadow: 0 10px 25px -5px rgba(0, 0, 0, 0.5);
50.     width: 100%;
51.     max-width: 480px;
52.     border: 1px solid #334155;
53. }
54.
55. h1 {
56.     font-size: 1.5rem;
57.     margin-bottom: 8px;
58.     color: var(--text-main);
59.     text-align: center;
60. }
61.
62. p.subtitle {
63.     font-size: 0.9rem;
64.     color: var(--text-muted);
65.     text-align: center;
66.     margin-bottom: 24px;
67. }
68.
69. .input-group {
70.     display: flex;
71.     flex-direction: column;
72.     gap: 12px;
73. }
74.
75. .password-wrapper {
76.     position: relative;
77.     width: 100%;
78.     display: flex;
79.     align-items: center;
80. }
81.
82. input[type="text"] {
83.     width: 100%;
84.     padding: 12px 60px 12px 16px;
85.     font-size: 1.1rem;
86.     letter-spacing: 1px;
87.     background-color: #0f172a;
```



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```
88.     color: var(--text-main);
89.     border: 2px solid #334155;
90.     border-radius: 8px;
91.     outline: none;
92.     transition: border-color 0.2s;
93. }
94.
95. input:focus {
96.     border-color: var(--primary);
97. }
98.
99. .toggle-btn {
100.     position: absolute;
101.     right: 12px;
102.     background: none;
103.     border: none;
104.     color: var(--text-muted);
105.     cursor: pointer;
106.     font-size: 0.85rem;
107.     padding: 4px 8px;
108.     border-radius: 4px;
109.     width: auto;
110. }
111.
112. .toggle-btn:hover {
113.     color: var(--text-main);
114.     background-color: transparent;
115. }
116.
117. button.btn-submit {
118.     width: 100%;
119.     padding: 12px;
120.     font-size: 1rem;
121.     font-weight: 600;
122.     color: #ffffff;
123.     background-color: var(--primary);
124.     border: none;
125.     border-radius: 8px;
126.     cursor: pointer;
127.     transition: background-color 0.2s;
128. }
129.
130. button.btn-submit:hover {
131.     background-color: var(--primary-hover);
132. }
133.
134. #result {
135.     margin-top: 20px;
```



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```
136.         padding: 14px 16px;
137.         border-radius: 8px;
138.         font-size: 0.95rem;
139.         text-align: center;
140.         display: none;
141.         word-break: break-word;
142.         border: 1px solid transparent;
143.     }
144.
145.     #result.warning {
146.         display: block;
147.         background-color: var(--warning-bg);
148.         color: var(--warning-text);
149.         border-color: var(--warning-border);
150.     }
151.
152.     #result.success {
153.         display: block;
154.         background-color: var(--success-bg);
155.         color: var(--success-text);
156.         border-color: var(--success-border);
157.     }
158.
159.     #result.error {
160.         display: block;
161.         background-color: var(--error-bg);
162.         color: var(--error-text);
163.         border-color: var(--error-border);
164.     }
165. </style>
166. </head>
167. <body>
168.
169.     <div class="container">
170.         <h1>Password Security Checker</h1>
171.         <p class="subtitle">Check if your password is a vulnerable
172.         palindrome</p>
173.
174.         <div class="input-group">
175.             <div class="password-wrapper">
176.                 <input type="text" id="passwordInput"
177.                 placeholder="Enter password..." />
178.                 <button type="button" class="toggle-btn" id="toggleBtn"
179.                 onclick="toggleVisibility()">Hide</button>
180.             </div>
181.             <button class="btn-submit"
182.             onclick="checkPassword()">Check Password</button>
183.         </div>
184.     </div>
```




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```
180.
181.     <div id="result"></div>
182. </div>
183.
184.     <script>
185.         let actualPassword = "";
186.         let isHidden = false;
187.
188.         const inputElement =
189.             document.getElementById("passwordInput");
190.         // Intercept input changes to support custom asterisk
191.         masking
192.         inputElement.addEventListener("input", function(e) {
193.             if (isHidden) {
194.                 // If characters were added or deleted
195.                 const diff = e.target.value.length -
196.                     actualPassword.length;
197.                 if (diff > 0) {
198.                     // Characters added
199.                     const addedChars =
200.                         e.target.value.slice(actualPassword.length);
201.                     actualPassword += addedChars;
202.                 } else if (diff < 0) {
203.                     // Characters removed (Backspace)
204.                     actualPassword = actualPassword.slice(0,
205.                         e.target.value.length);
206.                 }
207.                 // Render asterisks in the input display
208.                 inputElement.value = "*".repeat(actualPassword.length);
209.             } else {
210.                 actualPassword = e.target.value;
211.             }
212.         });
213.
214.         function toggleVisibility() {
215.             const btn = document.getElementById("toggleBtn");
216.
217.             if (!isHidden) {
218.                 // Switch to Hidden mode -> replace characters with *
219.                 isHidden = true;
220.                 inputElement.value = "*".repeat(actualPassword.length);
221.                 btn.textContent = "Show";
222.             } else {
223.                 // Switch to Visible mode -> show raw password
224.                 isHidden = false;
```



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```
223.         inputElement.value = actualPassword;
224.         btn.textContent = "Hide";
225.     }
226. }
227.
228.     function checkPassword() {
229.         const resultElement = document.getElementById("result");
230.         const password = actualPassword;
231.
232.         if (!password) {
233.             resultElement.className = "error";
234.             resultElement.textContent = "Please enter a password
first!";
235.             return;
236.         }
237.
238.         // Reverse exact password
239.         const reversedPassword =
password.split("").reverse().join("");
240.
241.         // Check if exact palindrome
242.         if (password === reversedPassword) {
243.             resultElement.className = "warning";
244.             resultElement.innerHTML = "⚠️ <strong>Weak
password!</strong> It is a palindrome. Choose a strong password.";
245.         } else {
246.             resultElement.className = "success";
247.             resultElement.innerHTML = "🔒 <strong>It is a strong
non-palindrome password.</strong>";
248.         }
249.     }
250.
251.     // Trigger check on Enter key press
252.     inputElement.addEventListener("keypress", function(event) {
253.         if (event.key === "Enter") {
254.             checkPassword();
255.         }
256.     });
257. </script>
258.
259. </body>
260. </html>
```



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3.Output:

The screenshot shows a web browser window with the address bar displaying "127.0.0.1:5500/Exp4/exp4_Aaryan_Kawale_1162CS.html". The page title is "Password Palindrome & Strength". The main content area features a "Password Security Checker" form. The form has a text input field containing "Aaryan" and a "Hide" button. Below the input field is a blue "Check Password" button. The output is displayed in a green box with a lock icon and the text: "It is a strong non-palindrome password."

The screenshot shows the same web browser window and application. The text input field now contains "1162-2611". The blue "Check Password" button is still present. The output is displayed in a brown box with a warning icon and the text: "Weak password! It is a palindrome. Choose a strong password."